Final Project Report — EV Market Intelligence

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1. Introduction — 1.1 Project Overview

The Global Electric Vehicle Market Intelligence Dashboard is designed to help users explore the EV market using available data on sales, vehicle categories, manufacturers, battery technologies, driving range, manufacturing country, and customer ratings.

1.2 Purpose

Provide a consistent interactive view for market strategy, competitive analysis, and sustainable-mobility research. The dashboard supports descriptive analysis; it does not establish causation or predict future sales unless separately validated.

2. Ideation Phase — 2.1 Problem Statement

EV industry data is distributed across many measures and categories. Emily Johnson needs category and manufacturer performance; David Chen needs market and production comparisons; Dr. Sarah Williams needs battery, range, geography, and satisfaction evidence.

2.2 Empathy Map Canvas

Users say they need reliable comparisons; think about data coverage and comparability; filter and compare views; feel pressure to make defensible decisions; experience pain from missing values and inconsistent labels; gain a consolidated, transparent dashboard.

2.3 Brainstorming

Prioritized ideas: annual sales trend, category and manufacturer ranking, battery technology distribution, driving-range comparison, country analysis, and customer-rating analysis. Priority is based on user value and feasibility.

3. Requirement Analysis — 3.1 Customer Journey Map

User opens the overview, reviews KPIs, selects a time period and category, compares manufacturers, explores battery/range/geography views, checks definitions and limitations, then exports or reports findings.

3.2 Solution Requirement

Functional: connect dataset, clean fields, calculate supported KPIs, build views, add filters, document source and refresh. Non-functional: usability, reliability, transparent calculations, responsive interaction, and secure handling of data.

3.3 Data Flow Diagram

Source dataset → data profiling and cleaning → validated measures and dimensions → Tableau worksheets → interactive dashboard/story → user exploration and reporting. Preserve raw data for reconciliation.

3.4 Technology Stack

Primary: Tableau Desktop or Tableau Public and a documented CSV/XLSX dataset. Optional: SQL/Python for preparation or Snowflake if required by the course. Use only tools actually used in the final implementation.

4. Project Design — 4.1 Problem–Solution Fit

A dashboard directly addresses the need to compare related market indicators in one interactive interface. Fit must be evaluated by verifying metric accuracy and gathering user feedback.

4.2 Proposed Solution

KPI cards; annual sales trend; category and manufacturer rankings; battery chemistry distribution; driving-range comparisons; manufacturing country view; customer ratings; filters and a guided story.

4.3 Solution Architecture

Data source → quality checks → calculated measures → Tableau worksheets → dashboard/story → validation and approved sharing. Data fields and chart choices depend on the actual dataset.

5. Project Planning & Scheduling — 5.1 Project Planning

Sprint 1: data audit and market overview. Sprint 2: battery, range, and geographic views. Sprint 3: ratings, story, and validation. Sprint 4: polish, documentation, and demonstration. Dates and completion status should be updated to match actual work.

6. Functional and Performance Testing — 6.1 Performance Testing

Test source-to-KPI reconciliation, filter behavior, date sorting, missing values, workbook reopen/refresh, and dashboard responsiveness. Record actual load times and results; no unmeasured performance claims are made.

7. Results — 7.1 Output Screenshots

Insert real screenshots of the overview dashboard, category/manufacturer analysis, battery/range comparison, geographic analysis, and filters after implementation. Do not insert mock screenshots as evidence.

8. Advantages & Disadvantages

Advantages: consolidated overview, interactive exploration, transparent metric definitions, and repeatable comparisons. Disadvantages: findings depend on dataset coverage, freshness, consistency, and availability of comparable ratings and market denominators.

9. Conclusion

The proposed dashboard provides a structured way to analyze EV market indicators and support evidence-based discussion. Final conclusions must be drawn from validated data and actual visual outputs.

10. Future Scope

Add scheduled refresh, regional market datasets, charging infrastructure, emissions/lifecycle metrics, time-series forecasting with validation, and user testing.

11. Appendix

Dataset link: to be inserted after selecting the actual source. Source code: Tableau calculated fields and any actual preparation scripts. GitHub/demo link: not yet available; add only a working link after publishing.

Evidence note: This is a project plan/documentation template. Add actual dataset source, screenshots, measured test results, and publication links only after completing and verifying the implementation.